

REVIEW

Subtyping of Primary Aldosteronism in Partial Adrenal Vein Sampling: The Relative Aldosterone Secretion Index

Federico Bernardo Rossi¹, Teresa Maria Seccia², Giacomo Rossitto³, Gian Paolo Rossi⁴

ABSTRACT: Expanded screening with the aldosterone-renin ratio has improved detection of primary aldosteronism, the most common surgically curable cause of arterial hypertension. However, identification of surgically curable primary aldosteronism remains constrained by technical limitations of subtyping by adrenal vein sampling (AVS), as bilateral selectivity often fails. Although alternative biomarkers superior to cortisol may help reduce this rate, currently, the failure to achieve bilateral selectivity precludes calculation of the lateralization index and, thus, diverts patients toward lifelong medical therapy. Recent advances have established the relative aldosterone secretion index (RASI) as a physiologically grounded strategy to interpret partially successful AVS. By quantifying aldosterone secretion from each adrenal gland relative to peripheral values and incorporating contralateral suppression, RASI-based interpretation can rescue many AVS studies by enabling subtyping under unilateral selectivity with 80% concordance with the lateralization index, when available. Postoperative outcomes following RASI-guided adrenalectomy approximate those achieved after fully selective AVS, with biochemical cure rates of 85% to 90% and blood pressure improvement in 65% to 75%. Studies published over the past 3 years have clarified factors influencing RASI performance, including cosyntropin stimulation, variations in aldosterone secretion, and the limited utility of cross-sectional imaging alone for subtype classification. Collectively, the available data support the incorporation of RASI into contemporary AVS interpretation algorithms. As primary aldosteronism management enters a postdetection era, subtyping has emerged as the principal bottleneck to definitive cure. RASI provides a pragmatic, evidence-based means to overcome the imperative dependence on bilateral selectivity while preserving diagnostic accuracy, thereby improving access to adrenalectomy and associated cardiovascular benefits.

Key Words: aldosterone ■ adrenal gland diseases sampling studies ■ blood pressure ■ critical pathways ■ diagnosis ■ hyperaldosteronism

Primary aldosteronism (PA) has undergone a conceptual transformation during the past decade. Once considered rare and esoteric, it is now recognized as the most common surgically curable form of arterial hypertension.^{1,2} Case detection has accelerated through broader use of the aldosterone-renin ratio and increased awareness^{3,4} of the cardiovascular burden of inadequately treated hyperaldosteronism.^{3,4} Consequently, the limiting step in the care pathway has shifted: the challenge is no longer to suspect and diagnose PA but to identify surgically curable unilateral forms and direct affected patients to adrenal ablation, the only intervention capable of achieving biochemical cure in over 90% of patients and substantial blood pressure improvement in most.^{5–7}

Despite promising results, functional imaging is applicable only in selected centers and in patients with PA with detectable adrenal nodules.⁸ Hence, adrenal vein sampling (AVS) remains the reference procedure for subtyping (Figure 1),⁹ and the only option in approximately one third of patients with confirmed PA, where imaging fails to detect adrenal nodules.¹⁰ However, AVS is inherently fragile: right adrenal cannulation is technically demanding, and failure to achieve bilateral selectivity precludes calculation of the lateralization index (LI), the established metric of unilateral aldosterone excess secretion.¹¹ Accordingly, using cortisol as a biomarker for selectivity, real-world cohorts demonstrate that 20% to 50% of AVS procedures fail to achieve bilateral selectivity.^{2,12,13} Although alternative selectivity

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Nonstandard Abbreviations and Acronyms	
AVS	adrenal vein sampling
LI	lateralization index
PA	primary aldosteronism
RASI	relative aldosterone secretion index

biomarkers, such as 17-OH progesterone and androstenedione, were reported to rescue many AVS studies judged to be nonselective using cortisol,¹⁴ the failure to achieve bilateral selectivity remains a major hurdle in the clinical management of patients with PA. Accordingly, surgically curable patients are frequently redirected toward lifelong mineralocorticoid receptor antagonist therapy or repeating AVS, which is costly and not widely available.¹⁵ This represents a structural bottleneck in the treatment pathway: surgically curable unilateral PA is lost not because the disease is undetectable, but because subtyping depends on binary procedural success requiring bilateral selectivity that is not always attainable (Figure 2).

PHYSIOLOGICAL BASIS AND DEVELOPMENT OF RELATIVE ALDOSTERONE SECRETION INDEX

The relative aldosterone secretion index (RASI)¹⁶ was proposed to address this vulnerability by quantifying the cortisol (C)-corrected aldosterone (A) ratio from each adrenal gland relative to peripheral values as follows:

A/C adrenal vein/A/C peripheral vein (=inferior vena cava).

This index is physiologically based on the Fick principle. Under the assumption that adrenal blood flow does not change substantially during AVS, the RASI estimates the aldosterone secretion rate, for example, the amount of aldosterone added to the arterial blood entering the adrenal gland (corresponding to plasma aldosterone concentration in a peripheral vein or the infra-adrenal inferior vena cava) corrected for the degree of adrenal vein catheterization selectivity (Figure 1).¹⁷ Grounded on the premise that excess aldosterone from a dominant adrenal, by increasing blood volume and pressure, suppresses renin, angiotensin II, and, thereby, aldosterone production, RASI also estimates suppression of hormone secretion from the contralateral gland.¹⁷

Thus, RASI is applicable to both sides but requires selective AVS only on one side. Mathematically, it is equal to the contralateral suppression index in the non-dominant side; however, by definition, the latter requires bilateral catheterization to be identified. Under stimulated or unstimulated conditions, a high RASI indicates dominant aldosterone secretion, whereas a low RASI denotes suppression.

The subsequent development of pragmatic thresholds, RASI ≥3 on the dominant side and <1 contralaterally, which were established in unstimulated studies, enabled clinical application in partially successful AVS, allowing interpretation under unilateral selectivity.¹⁸ RASI, thus, reframes the diagnostic question from bilateral comparison to gland-specific secretion relative to systemic context, consistent with the biology of unilateral hypersecretion.

CLINICAL VALIDATION AND DIAGNOSTIC PERFORMANCE

We searched PubMed/MEDLINE and Embase from January 2018 to March 2026 using the terms primary aldosteronism, adrenal vein sampling, and RASI or contralateral suppression, restricted to original articles in human subjects with confirmed PA.

The relevant studies show that since its introduction,¹⁶ RASI has been validated across independent cohorts¹⁶ (Table 1). In a large multicenter international study, RASI demonstrated the skewed distribution of values between surgically cured unilateral PA and bilateral PA, thus proving the feasibility of identifying unilateral PA without bilateral cannulation success and providing a clinical precedent for RASI-based decision-making. However, it showed overlap with presumed bilateral PA,^{18,22} particularly with the use of restrictive thresholds.^{22,23} Importantly, patients with PA classified as bilateral,²⁴ including those with elevated nondominant RASI but asymmetrical hypersecretion, have shown favorable biochemical and clinical outcomes after unilateral adrenal ablation.²⁵

Multiple single-center and multicenter series have operationalized RASI under unilateral selectivity using various cutoffs. In consecutive patients undergoing cosyntropin-stimulated AVS, bilateral selectivity failed in 17%, yet a nondominant RASI <0.5 demonstrated 87% sensitivity and 93% specificity for predicting contralateral lateralization.¹² Similar skewed RASI distributions between unilateral and bilateral PA have been reported across multiple additional cohorts.^{26–31}

Collectively, these studies indicate that RASI can rescue ≈20% to 50% of incomplete AVS procedures that would otherwise remain nondiagnostic,¹⁵ with concordance exceeding 80% relative to LI when LI is calculable. Postoperative biochemical cure approaches 85% to 90%, with clinical blood pressure improvement observed in 65% to 75%, approximating outcomes achieved after bilaterally selective AVS (Figure 2).

On the other hand, reports describing overlap of RASI values between unilateral and bilateral PAs should be interpreted with caution because of the stringent lateralization frameworks that lead to the classification of asymmetrical aldosterone hypersecretion as

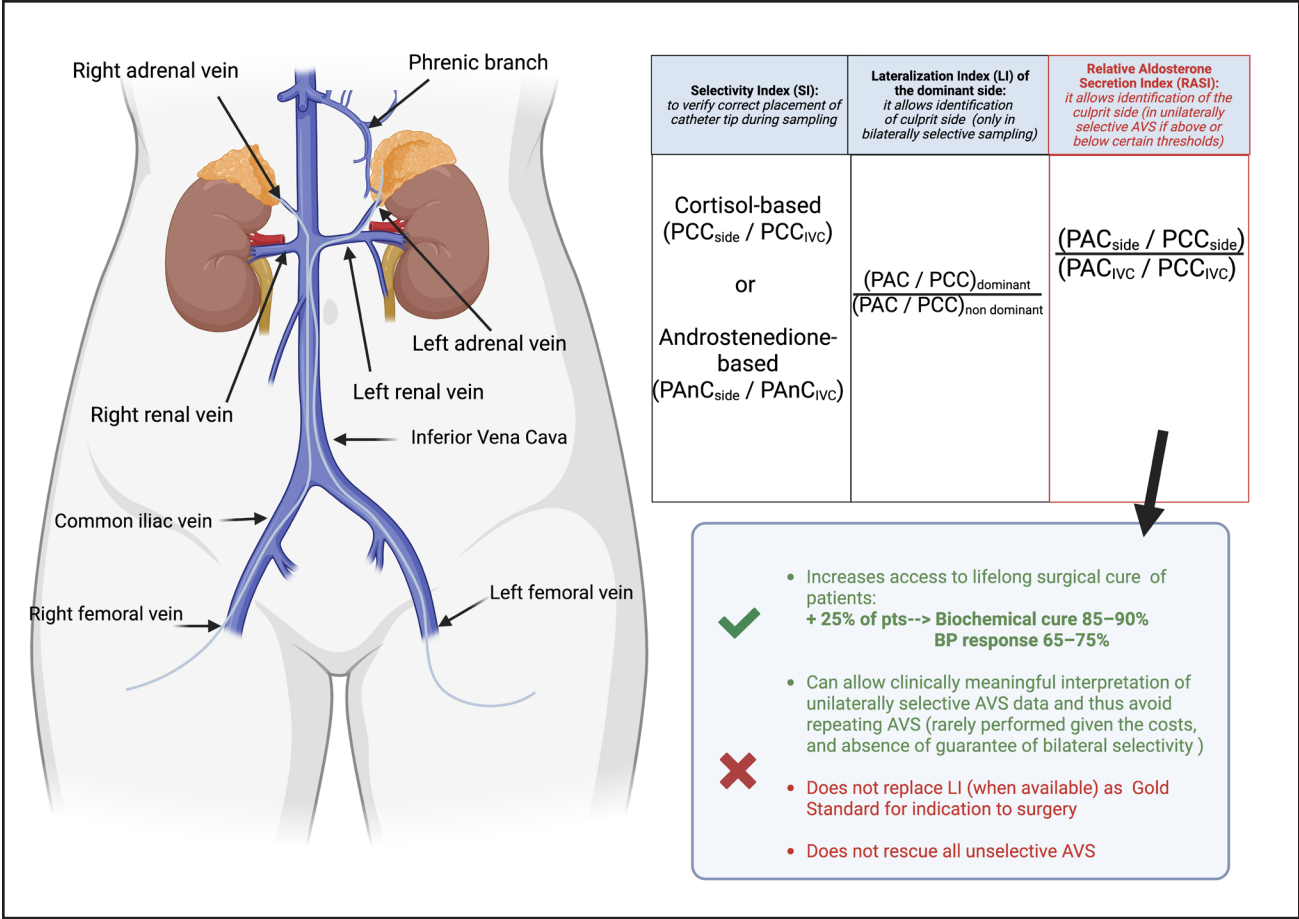


Figure 1. The cartoon illustrates the procedure of adrenal vein sampling (AVS) and the indices used for its interpretation, incorporating the relative aldosterone secretion index (RASI).

Note that: the contralateral suppression index used as supplemental criteria in bilaterally selective AVS is mathematically identical to the RASI of the nondominant side. Created in BioRender. Rossi, F.B. (2026) <https://BioRender.com/ur0vx9m>. BP indicates blood pressure; Dominant, side with higher plasma aldosterone concentration; IVC, inferior vena cava; Nondominant, contralateral side; PAC, plasma aldosterone concentration; PAnC, plasma androstenedione concentration; and PCC, plasma cortisol concentration.

bilateral disease.^{22,23,32} The methodological choice of applying restrictive lateralization and bilateral selectivity thresholds leads to reassigning biologically unilateral cases, including those with dominant secretion and contralateral suppression, to the bilateral category, thus artificially inflating apparent RASI overlap and underestimating its discriminatory performance. In contrast, interpretation strategies incorporating unilateral selectivity and aldosterone excess or contralateral suppression, based on more lenient thresholds,³³ align more closely with adrenal physiology and preserve access to surgical cure.³³

OPERATIONAL ADVANTAGES IN PARTIAL AVS

Complementary observations further demonstrate the potential of RASI to salvage right-sided cannulation failures, which is the most common cause of incomplete bilateral selectivity.

In cohorts with right adrenal lesions, about 30% of which experienced failed right sampling, contralateral suppression guided surgical decision-making, leading, in all cases, to histopathology-confirmed aldosterone-producing adenomas, biochemical cure, and improved blood pressure control postoperatively.³⁴

Collectively, these data illustrate that incomplete AVS need not preclude cure: subtyping can be recovered when secretion is interpreted against peripheral context,^{26–31} rather than exclusively through bilateral comparison. They also suggest that left catheterization and RASI can be useful to identify the culprit side, particularly in centers with a high rate of failure on the right-hand side.

DETERMINANTS OF RASI AND CAVEATS ON ITS USE

As with all tests, the RASI is not perfectly accurate, and important boundary conditions remain (Table 2).

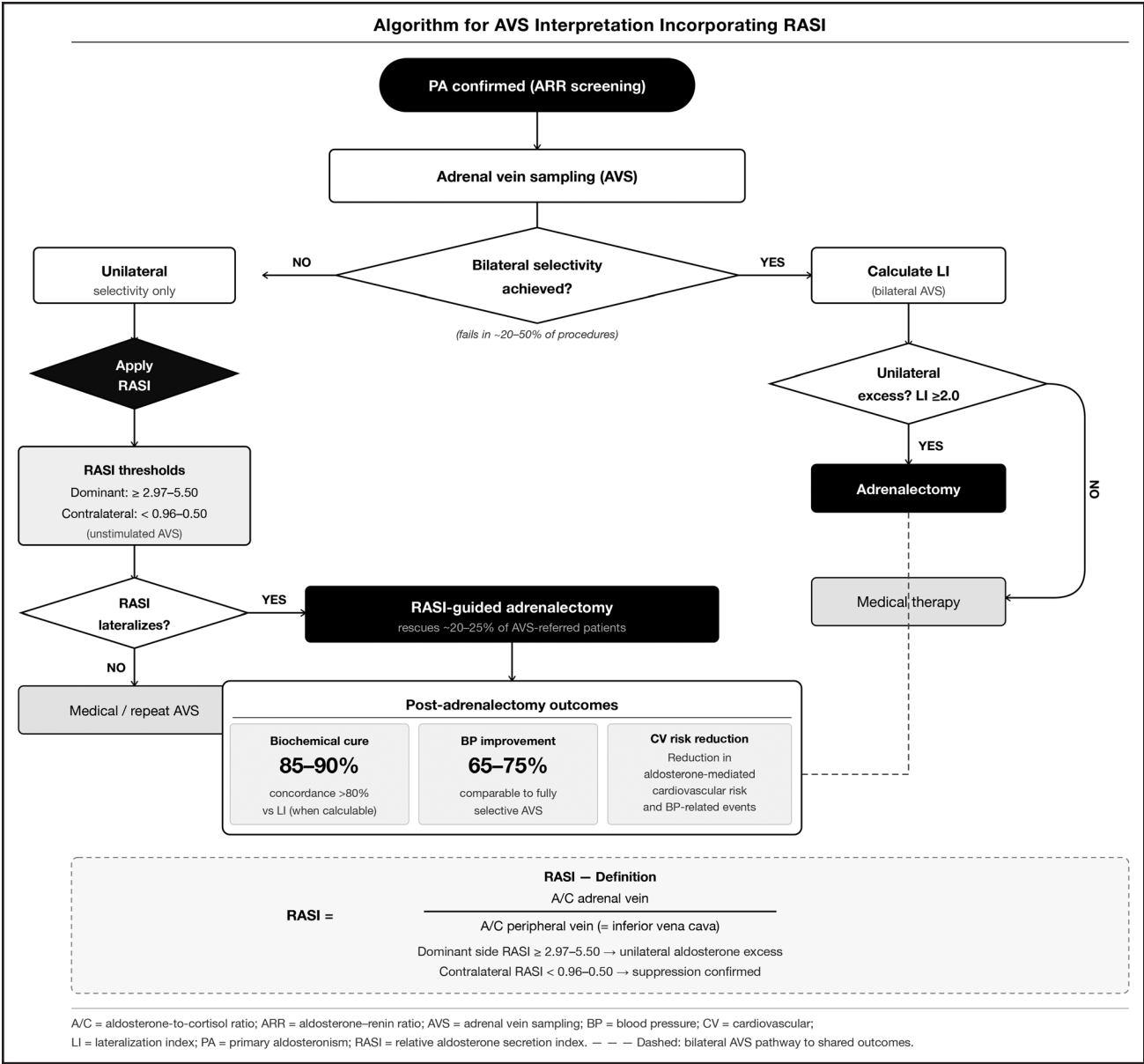


Figure 2. Algorithm for adrenal vein sampling (AVS) interpretation incorporating relative aldosterone secretion index (RASI). Proposed workflow for subtyping primary aldosteronism. Incomplete bilateral selectivity occurs in ≈20% to 50% of AVS procedures even at referral centers. Under unilateral selectivity, which is achieved in close to 100% of the patients, application of the relative aldosterone secretion index (RASI; dominant ≥2.97–5.50 and contralateral <0.96–0.50, based on unstimulated AVS) enables identification of unilateral disease. RASI-based interpretation rescues ≈20% to 50% of AVS-referred patients, avoiding the need for repeated AVS, which is costly and hardly feasible, and preserving access to adrenalectomy with high rates of biochemical cure and blood pressure (BP) response. Summary of outcomes across validation cohorts showed concordance >80% relative to lateralization index when calculable, biochemical cure rates of 85% to 90%, and clinical BP improvement in 65% to 75% of patients undergoing RASI-guided adrenalectomy, translating into improved blood pressure control and reduction in aldosterone-mediated cardiovascular risk.

However, studies involving >1000 patients with PA (Table 1) globally have consolidated these observations and expanded external validation across centers, reinforcing RASI as a reproducible adjunct to conventional AVS interpretation.^{18,29,35,36}

Mechanistic analyses have also identified some factors influencing RASI discrimination. For example, aldosterone secretion is variable and has been reported to fall to normal values even during the short time required for the

procedure.³⁷ Caution should, therefore, be exercised in using the RASI when aldosterone secretion is at a nadir, as evidenced by normal plasma aldosterone concentration.^{10,22,37,38} Moreover, RASI cannot be used when AVS is unsuccessful bilaterally because it requires successful unilateral catheterization as assessed with reliable cortisol normalization and careful peripheral sampling.^{10,22,37,38} adrenocortrophic hormone (cosyntropin) stimulation, which increases cortisol (more than aldosterone),

Table 1. Studies Evaluating RASI (CLSI) in AVS Interpretation

Study/year	No. of patients with PA/study type/country	Reference index	CLSI used	RASI used	Was CLSI or RASI useful?	Key limitations
Kline/2015 ¹⁹	99/single center/Canada	LI; HTN outcome	Yes	No	Yes: predicted lateralization and BP outcome	Single center; retrospective; 43% on either ACE inhibitor or ARB, 27% using a diuretic; better prediction postcosyntropin
Wang/2018 ²⁹	222/multicenter-center/US, Canada	LI (bilateral AVS)	No	No	No: incomplete AVS unreliable	Only the left RASI used. No surgical gold standard
Burrello/2020 ⁶	158/single center/Italy	Outcome-based subtype	Yes	No	Yes: contralateral suppression predictive	Retrospective; ML-derived
Yang/2021 ²⁰	263/multicenter/South Korea	LI; PASO outcome	Yes	No	Limited discrimination (CLS prevalent in 94%)	ACTH-stimulated; reduced specificity; CLSI cut-off <0.26 in many cases, suggesting a denominator effect due to cortisol normalization
Rossi/2023 ¹⁸	1625/multicenter/international	Outcome-based (surgical cure)	Yes	Yes	Yes: strong discrimination vs surgical gold standard	Protocol heterogeneity
Chow/2024 ²⁰	50/single center/China	LI and outcome	Yes	Yes	Yes: improved PPV with combined thresholds	Low-volume center
Grundy/2025 ²²	29/single center/Canada	Outcome (PASO)	Yes	No	Yes: biochemical improvement despite asymmetry	Selected asymmetrical bilateral; ACTH included
Basson/2026 ²¹	362/single center/Canada	LI ≥4 (simulation)	Yes	No	Yes: improved PPV	Simulation model; LI-based reference
Vibhatavata/2026 ¹⁹	≈200/single center/United States	LI reference	Yes	No	Limited utility; substantial overlap between subtypes	LI-based reference; no outcome validation

ACE indicates angiotensin-converting enzyme; ACTH, adrenocorticotrophic hormone; ARB, angiotensin receptor blockers; AVS, adrenal vein sampling; BP, blood pressure; CLS, contralateral suppression; CLSI, contralateral suppression index; HTN, hypertension; LI, lateralization index; ML, machine learning; PA, primary aldosteronism; PASO, Primary Aldosteronism Surgical Outcome criteria; PPV, positive predictive value; and RASI, relative aldosterone secretion index.

enhances confidence in catheterization success^{33,39} but reduces interglandular aldosterone gradients. By increasing the denominators in the ratio more than the numerator, it decreases RASI on the culprit side. Accordingly, it lowered the LI in 58% of the patients and may lead to the classification of unilateral PA as bilateral.⁴⁰ Moreover, it can cause factitious suppression contralaterally due to this denominator effect.³³ Thus, it should be avoided when unstimulated sampling is feasible.¹⁷

Bilateral micronodular hyperplasia, cortisol-cosecretion, chronic kidney disease, and inadvertent sampling of the accessory hepatic⁴¹ on the right-hand side, or of the renal vein distal to the adrenal vein on the left-hand side, may complicate interpretation.

Imaging remains important to exclude carcinoma and contextualize venous findings.²² Accordingly, cross-sectional imaging features have been proposed to improve the diagnostic performance of the RASI.^{22,38} Unfortunately, their usefulness is limited in ≈41% of the patients with PA, who have negative imaging (34%) and bilateral adrenal nodules (7%).¹⁰

IMPLICATIONS FOR CONTEMPORARY PA CARE

Viewed collectively, these developments support a reordering of priorities in PA care. The usefulness of confirmatory suppression testing to identify unilateral PA and avoid AVS is debated^{42,43} and is no longer considered universally necessary or cost-effective in florid disease in

the latest guidelines,⁹ based on results of 2 large meta-analyses, a recent study,^{44–46} and the contrasting findings obtained even with the seated saline suppression test.^{47–49} Concurrently, increased aldosterone-renin ratio–based screening has highlighted substantial underrecognition of PA in hypertension clinics.^{50,51}

Within this evolving continuum, AVS has emerged as the rate-limiting step, not solely because of its limited availability, but because the requirement for bilateral success determines access to cure. Incorporation of RASI to assess dominant secretion and contralateral suppression might overcome this constraint (Figure 2) without compromising diagnostic fidelity and avoiding repeating AVS, which is costly and unfeasible in many conditions. Confidence in clinical decision-making increases in parallel with raising and decreasing RASI values in the dominant and contralateral sides, respectively.^{18,26}

Accordingly, unilateral selectivity combined with RASI may provide supportive evidence of unilateral PA in appropriately selected patients, in whom peripheral sampling is available. This approach does not diminish the value of bilateral selectivity and LI when achievable, nor does it advocate indiscriminate surgery. Rather, it addresses loss of curative opportunity driven by AVS procedural technical fragility.

CLINICAL OUTCOMES AND IMPLICATIONS

The clinical implications of RASI are broad and substantial. The distinction between surgical cure and chronic

Table 2. Determinants of RASI Performance and Caveats in Interpretation

Confounder/determinant	Effect on RASI	Direction of bias
Unreliable unilateral catheterization	Invalid cortisol normalization; inaccurate RASI	Unpredictable/misclassification
Inaccurate peripheral (IVC) sampling	Distorts denominator; misestimates secretion	↑ or ↓ RASI
Aldosterone secretion at the nadir	Underestimates dominant secretion	↓ RASI (false bilateral)
Physiological aldosterone variability during AVS	Transient normalization may mask lateralization.	↓ RASI
ACTH (cosyntropin) stimulation	Reduces intraglandular gradient	↓ dominant RASI
ACTH denominator effect (↑ cortisol)	Expands denominator; lowers ratio	↓ RASI
Factitious contralateral suppression under ACTH	Artificial nondominant suppression	Artificial ↓ non-dominant RASI
Bilateral micronodular hyperplasia	Blunts asymmetry	Reduced discrimination
Cortisol cosecretion	Alters cortisol normalization	Unpredictable distortion
Chronic kidney disease	Alters hormone kinetics	Uncertain bias
Accessory hepatic vein sampling (right)	Falsely low aldosterone (dilution)	↓ RASI
Renal vein sampling distal to adrenal (left)	Dilution artifact	↓ RASI
Negative cross-sectional imaging	Limits contextual interpretation	No direct ratio effect
Bilateral adrenal nodules on imaging	Reduces the specificity of lateralization	No direct ratio effect

ACTH indicates adrenocorticotrophic hormone; IVC, inferior vena cava; and RASI, relative aldosterone secretion index.

medical therapy defines divergent long-term cardiovascular trajectories. Compared with medical therapy alone, in unilateral PA, adrenalectomy reduces cardiovascular risk beyond blood pressure control, particularly regarding cardiovascular damage,⁵ atrial fibrillation,⁵² and metabolic sequelae of aldosterone excess. Therefore, AVS is important for the management of asymmetrical bilateral PA in selected patients, particularly those in whom medical therapy is inadequate, poorly tolerated, or not feasible. Debulking with unilateral adrenalectomy provides favorable early clinical outcomes, even when the RASI value in the nondominant side, for example, the contralateral suppression index, showed no contralateral suppression.^{25,53} Prospective multicenter validation with standardized RASI reporting and established clinical and biochemical outcome criteria will further define generalizability.

CONCLUSIONS

As screening expands, PA has entered a postdetection era. The principal challenge is, therefore, to ensure timely access to curative pathways based on adrenal ablation in patients with unilateral PA.⁵⁴ The use of RASI

is associated with favorable biochemical and clinical outcomes after unilateral adrenal ablation even when computed on the nondominant side.¹⁸ By enabling the interpretation of partially successful AVS, RASI directly addresses this bottleneck. Integration of RASI into diagnostic algorithms (Figure 2) would strengthen continuity between detection and cure, align practice with physiology, and reduce unnecessary diversion to lifelong medical treatment in surgically curable PA.

ARTICLE INFORMATION

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Disclosures

None.

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